

Practical Code and Output Report

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Practical 2

Code: practical2 AWD.html

```
HTML
<div ng-app="orderApp" ng-controller="orderCtrl">
  <h2>Product Order Form</h2>
  <div ng-repeat="item in products">
    {{item.name}} (Price: {{item.price}}):
    <input type="number" ng-model="item.quantity" placeholder="Enter quantity" title="Item quantity">
  </div>
  <hr>
  <p>Bill Amount: {{billAmount | number:2}}</p>
  <p>GST (18%): {{gst | number:2}}</p>
  <p><strong>Total Amount: {{total | number:2}}</strong></p>
  <p>Made By Narendra 09</p>
</div>

<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
<script>
  var app = angular.module('orderApp', []);
  app.controller('orderCtrl', function($scope) {
    $scope.products = [
      {name: 'Laptop', price: 50000, quantity: 0},
      {name: 'Mouse', price: 500, quantity: 0},
      {name: 'Keyboard', price: 1200, quantity: 0},
      {name: 'Monitor', price: 8000, quantity: 0},
      {name: 'USB Cable', price: 300, quantity: 0}
    ];
    $scope.calculate = function() {
      $scope.billAmount = 0;
      $scope.products.forEach(p => {
        $scope.billAmount += (p.price * p.quantity);
      });
      $scope.gst = $scope.billAmount * 0.18;
      $scope.total = $scope.billAmount + $scope.gst;
    };
  });
</script>
```

Output

HTML

Product Order Form

Laptop (Price: 50000):

Mouse (Price: 500):

Keyboard (Price: 1200):

Monitor (Price: 8000):

USB Cable (Price: 300):

Bill Amount:

GST (18%):

Total Amount:

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Practical 3

Code: practical3 AWD.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>AngularJS Prime Filter</title>
  <link rel="stylesheet" href="styles.css">
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
</head>
<body ng-app="myApp" ng-controller="myCtrl">

  <div class="prime-checker-container">
    <h2>Prime Checker</h2>
    <label>Enter a Number:</label>
    <input type="number" ng-model="userNumber" placeholder="Type a number...">

    <hr>

    <p><strong>Result:</strong> {{ userNumber | isPrime }}</p>
    <p>Made By Narendra 09</p>
  </div>

  <script>
    var app = angular.module('myApp', []);

    // Custom filter logic
    app.filter('isPrime', function() {
      return function(input) {
        if (!input || input < 2) {
          return "Enter a number greater than 1";
        }

        var isPrime = true;
        // Standard prime check algorithm
        for (var i = 2; i <= Math.sqrt(input); i++) {
          if (input % i === 0) {
            isPrime = false;
            break;
          }
        }

        return isPrime ? input + " is a Prime Number" : input + " is NOT a Prime Number";
      };
    });

    app.controller('myCtrl', function($scope) {
      // Controller can remain empty for this logic as the filter handles the processing
    });
  </script>

</body>
</html>
```

Output

Prime Checker

Enter a Number:

Result: 17 is a Prime Number

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Practical 4

Code: practical4 AWD.html

```
<!DOCTYPE html>
<html lang="en" ng-app="squareApp">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>AngularJS Square Service</title>
<link rel="stylesheet" href="styles.css">
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
</head>
<body ng-controller="squareCtrl">

    <h2>Assignment 4: Square Calculation using Service</h2>
    <hr>

    <label>Enter a Number:</label>
    <input type="number" ng-model="inputNum" placeholder="e.g. 5">

    <button ng-click="findSquare()">Calculate Square</button>

    <div class="result-container">
        <strong ng-if="result">The Square of {{inputNum}} is: {{result}}</strong>
        <p>Made by Narendra 09.</p>
    </div>

    <script>
        // 1. Define the Application Module
        var app = angular.module('squareApp', []);

        // 2. Define the Custom Service
        app.service('calcService', function() {
            this.square = function(x) {
                return x * x;
            };
        });

        // 3. Define the Controller and Inject the Service
        app.controller('squareCtrl', function($scope, calcService) {
            $scope.findSquare = function() {
                if ($scope.inputNum !== undefined) {
                    // Call the square function from the injected service
                    $scope.result = calcService.square($scope.inputNum);
                } else {
                    alert("Please enter a valid number");
                }
            };
        });
    </script>

</body>
</html>
```

Output

Assignment 4: Square Calculation using Service

Enter a Number:

The Square of 8 is: 64

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Practical 5

Code: practical5 AWD.html

```
<!DOCTYPE html>
<html ng-app="calcApp" lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>AngularJS Service - Add/Sub</title>
  <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
  <style>
    body { font-family: Arial, sans-serif; padding: 20px; }
    .container { border: 1px solid #ccc; padding: 20px; width: 300px; border-radius: 8px; }
    input { margin-bottom: 10px; width: 90%; padding: 5px; }
    button { margin-right: 5px; padding: 5px 15px; cursor: pointer; }
  </style>
</head>
<body ng-controller="calcCtrl">

<div class="container">
  <h3>Arithmetic Service</h3>

  <label>Number 1:</label><br>
  <input type="number" ng-model="num1" placeholder="Enter first number"><br>

  <label>Number 2:</label><br>
  <input type="number" ng-model="num2" placeholder="Enter second number"><br>

  <button ng-click="doAddition()">Add</button>
  <button ng-click="doSubtraction()">Sub</button>

  <hr>
  <h4>Result: {{ result }}</h4>
  <p>Made by Narendra 09.</p>
</div>

<script>
  // 1. Define the Module
  var app = angular.module('calcApp', []);

  // 2. Define the Service
  app.service('MathService', function() {
    this.add = function(a, b) {
      return (parseFloat(a) || 0) + (parseFloat(b) || 0);
    };

    this.subtract = function(a, b) {
      return (parseFloat(a) || 0) - (parseFloat(b) || 0);
    };
  });

  // 3. Define the Controller and Inject the Service
  app.controller('calcCtrl', function($scope, MathService) {
    $scope.num1 = 0;
    $scope.num2 = 0;
    $scope.result = 0;

    $scope.doAddition = function() {
      $scope.result = MathService.add($scope.num1, $scope.num2);
    };
  });
</script>
</body>
</html>
```



```
$scope.doSubtraction = function() {  
    $scope.result = MathService.subtract($scope.num1, $scope.num2);  
};  
});  
</script>  
  
</body>  
</html>
```

Output

Arithmetic Service

Number 1:

Number 2:

Result: 35

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Practical 6

Code: practical6 AWD.js

```
// Import the built-in HTTP module
const http = require('http');
// Create the server
const server = http.createServer((req, res) => {
  // Set the response header to HTML
  res.setHeader('Content-Type', 'text/html');

  // Get the URL from the request object
  const url = req.url;

  // Routing logic based on URL
  if (url === '/' || url === '/home') {
    res.statusCode = 200;
    res.write('<h1>Welcome to the Home Page</h1>');
    res.write('<p>This is the main content of our Node.js server. Developed by Narendra 09.</p>');
  }
  else if (url === '/about') {
    res.statusCode = 200;
    res.write('<h1>About Us</h1>');
    res.write('<p>We are learning Advanced Web Development (MCA).</p>');
  }
  else if (url === '/contact') {
    res.statusCode = 200;
    res.write('<h1>Contact Page</h1>');
    res.write('<p>Email us at: student@university.edu</p>');
  }
  else {
    // Handling 404 - Page Not Found
    res.statusCode = 404;
    res.write('<h1 style="color: red;">404 - Page Not Found</h1>');
    res.write('<p>The URL you requested does not exist.</p>');
  }

  // End the response
  res.end();
});

// Server listens on port 3000
const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server is running at http://localhost:${PORT}`);
  console.log('Try visiting /about or /contact in your browser.');
  console.log('Made by Narendra 09.');
});
```

Output

Welcome to the Home Page

This is the main content of our Node.js server. Developed by Narendra 09.

Practical 7

Code: practical7 AWD.js

```
const mongoose = require('mongoose');

// --- CONNECTION ---
mongoose.connect('mongodb://127.0.0.1:27017/AWD_Lab_DB')
  .then(() => {
    console.log('Connected to MongoDB...');
    createDataModels(); // ✅ CALL AFTER CONNECTION
  })
  .catch(err => console.error('Could not connect...', err));

// --- EMBEDDED MODEL ---
const addressSchema = new mongoose.Schema({
  street: String,
  city: String,
  zip: Number
});

const studentSchema = new mongoose.Schema({
  name: String,
  age: Number,
  address: addressSchema
});

const Student = mongoose.model('Student', studentSchema);

// --- REFERENCE MODEL ---
const teacherSchema = new mongoose.Schema({
  name: String,
  department: String
});

const Teacher = mongoose.model('Teacher', teacherSchema);

const courseSchema = new mongoose.Schema({
  title: String,
  credits: Number,
  instructor: {
    type: mongoose.Schema.Types.ObjectId,
    ref: 'Teacher'
  }
});

const Course = mongoose.model('Course', courseSchema);

// --- FUNCTION ---
async function createDataModels() {
  try {
    // Embedded
    const student = new Student({
      name: "Narendra Borhade",
      age: 9,
      address: { street: "Charoli phata", city: "Pune", zip: 412105 }
    });

    await student.save();
    console.log("Embedded Student Record Created");

    // Referenced
```

```
const teacher = new Teacher({
  name: "Dr. Priynka Abhang",
  department: "MCA"
});

await teacher.save();

const course = new Course({
  title: "Advanced Web Development",
  credits: 4,
  instructor: teacher._id
});

await course.save();
console.log("Referenced Course Record Created");

} catch (err) {
  console.error("Error:", err);
} finally {
  mongoose.disconnect();
}
}
```

Output

practical7 AWD.js Console Output

```
Connected to MongoDB...
Embedded Student Record Created
Referenced Course Record Created
```

Practical 8

Code: practical8 AWD.js

```
const mongoose = require('mongoose');
mongoose.connect('mongodb://localhost:27017/schoolDB');
const studentSchema = new mongoose.Schema({ name: String, rollNo: Number });
const Student = mongoose.model('Student', studentSchema);
// Create Example
const newStudent = new Student({ name: "Narendra Borhade", rollNo: 9 });
newStudent.save().then(() => console.log("Record Inserted"));
// Read Example
Student.find().then(data => console.log(data));
```

Output

practical8 AWD.js Console Output

```
Record Inserted
[
  {
    _id: new ObjectId('69bd7fde2dfbc19b61b42e23'),
    name: 'John Doe',
    rollNo: 101,
    __v: 0
  },
  {
    _id: new ObjectId('69bd800fcec80f87348000b3'),
    name: 'John Doe',
    rollNo: 101,
    __v: 0
  },
  {
    _id: new ObjectId('69c27964bbdfaa3d759a71d7'),
    name: 'John Doe',
    rollNo: 101,
    __v: 0
  },
  {
    _id: new ObjectId('69c279ba7fc01d016068747c'),
    name: 'Narendra Borhade',
    rollNo: 9,
    __v: 0
  },
  {
    _id: new ObjectId('69c2bdab76f9d193e75a77bc'),
    name: 'Narendra Borhade',
    rollNo: 9,
    __v: 0
  },
  {
    _id: new ObjectId('69ea226d78e7805713f54e91'),
    name: 'Narendra Borhade',
    rollNo: 9,
    __v: 0
  },
  {
    _id: new ObjectId('69ea2658f151756940801b4f'),
    name: 'Narendra Borhade',
    rollNo: 9,
    __v: 0
  },
  {
    _id: new ObjectId('69ea26a3e70b0184a00ea01c'),
    name: 'Narendra Borhade',
    rollNo: 9,
    __v: 0
  },
  {
    _id: new ObjectId('69ea271cbb024370cf27e1ef'),
    name: 'Narendra Borhade',
    rollNo: 9,
    __v: 0
  }
]
```

Practical 9

Code: practical9 AWD.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Records Management</title>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
<style>
body { font-family: Arial, sans-serif; margin: 20px; background-color: #f4f4f4; }
.container { background: white; padding: 20px; border-radius: 8px; box-shadow: 0 0 10px rgba(0,0,0,0.1); }
table { width: 100%; border-collapse: collapse; margin-top: 20px; } th, td { border: 1px solid #ddd; padding: 10px; text-align: left; }

th { background-color: #007bff; color: white; }
.btn-add { background: #28a745; color: white; padding: 10px; border: none; cursor: pointer; }
.btn-delete { background: #dc3545; color: white; padding: 5px; border: none; cursor: pointer; }
</style>
</head>
<body ng-app="studentApp" ng-controller="studentCtrl">

<div class="container">
<h2>Manage Student Records</h2>

<fieldset>
<legend>Add New Student</legend>
<input type="text" ng-model="newStudent.name" placeholder="Enter Name">
<input type="number" ng-model="newStudent.roll" placeholder="Enter Roll No">
<input type="text" ng-model="newStudent.dept" placeholder="Enter Department">
<button class="btn-add" ng-click="addRecord()">Add Student</button>
</fieldset>

<table>
<thead>
<tr>
<th>Name</th>
<th>Roll No</th>
<th>Department</th>
<th>Action</th>
</tr>
</thead>
<tbody>
<tr ng-repeat="student in students">
<td>{{ student.name }}</td>
<td>{{ student.roll }}</td>
<td>{{ student.dept }}</td>
<td>

<button class="btn-delete" ng-click="deleteRecord($index)">Remove</button>
</td>
</tr>
</tbody>
</table>
<p ng-if="students.length == 0">No records found. Add a student to begin.</p>
</div>

<script>
var app = angular.module('studentApp', []); app.controller('studentCtrl', function($scope) {
// Initial Data
$scope.students = [
```

```

{ name: "Narendra Borhade", roll: 9, dept: "MCA" },
{ name: "Vaibhav Bhor", roll: 7, dept: "MCA" },
{ name: "Vaibhav singh", roll: 33, dept: "BCA" }
];

// Function to Add Record
$scope.addRecord = function() {
if($scope.newStudent.name && $scope.newStudent.roll) {
$scope.students.push(angular.copy($scope.newStudent));
// Clear input fields
$scope.newStudent = {};
} else {
alert("Please fill in Name and Roll Number.");
}
};

// Function to Delete Record
$scope.deleteRecord = function(index) {
if(confirm("Are you sure you want to delete this record?")) {
$scope.students.splice(index, 1);
}
};

});
</script>

</body>
</html>

```

Output

Manage Student Records

Add New Student

Name	Roll No	Department	Action
Narendra Borhade	9	MCA	Remove
Vaibhav Bhor	7	MCA	Remove
Vaibhav singh	33	BCA	Remove
Test Student	101	MCA	Remove

Practical 10

Code: practical10 AWD.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Add Student Record</title>
</head>
<body>
<h2>Student Registration Form</h2>
<form action="/add-student" method="POST">
<label>Name:</label><br>
<input type="text" name="name" placeholder="Enter your full name" required><br><br>

<label>Email:</label><br>
<input type="email" name="email" placeholder="Enter your email address" required><br><br>

<label>Course:</label><br>
<input type="text" name="course" placeholder="Enter your course name" required><br><br>

<button type="submit">Submit to MongoDB</button>
</form>
</body>
</html>
```

Code: practical10 AWD1.js

```
const express = require('express');
const mongoose = require('mongoose');
const path = require('path');

const app = express();

// Middleware to parse form data (using Express's built-in parser)
app.use(express.urlencoded({ extended: true }));

// MongoDB Connection
mongoose.connect('mongodb://localhost:27017/labDB')
  .then(() => console.log("Connected to MongoDB..."))
  .catch(err => console.error("Connection failed", err));

// Define Schema and Model
const studentSchema = new mongoose.Schema({
  name: String,
  email: String,
  course: String
});
const Student = mongoose.model('Student', studentSchema);

// Route to serve the HTML Form
app.get('/', (req, res) => {
  // FIXED: Added .html extension to the file name
  res.sendFile(path.join(__dirname, 'practical10 AWD.html'));
});

// Route to handle Form Submission
app.post('/add-student', async (req, res) => {
  const newStudent = new Student({
    name: req.body.name,
    email: req.body.email,
```

```
        course: req.body.course
    });

    try {
        await newStudent.save();
        res.send("<h2>Data successfully inserted into MongoDB!</h2><a href='/'>Go Back</a>");
    } catch (err) {
        res.status(500).send("Error inserting data: " + err);
    }
});

app.listen(3000, () => console.log("Server started on http://localhost:3000"));
```

Output

Data successfully inserted into MongoDB!

[Go Back](#)
